21/11/2025

Z- Transform

Given a discrete-time signal $x[n]$,

the z- transform is defined as -

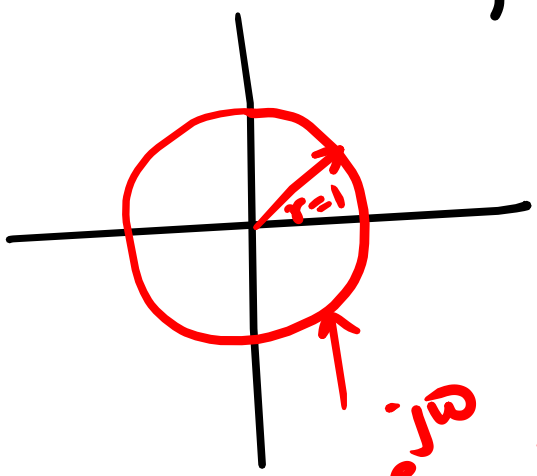
$$\boxed{X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}}$$

if $X(z) = \infty$ for some z, it is a pole

if $z = e^{j\omega}$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \Leftarrow \text{D.T.F.T.}$$

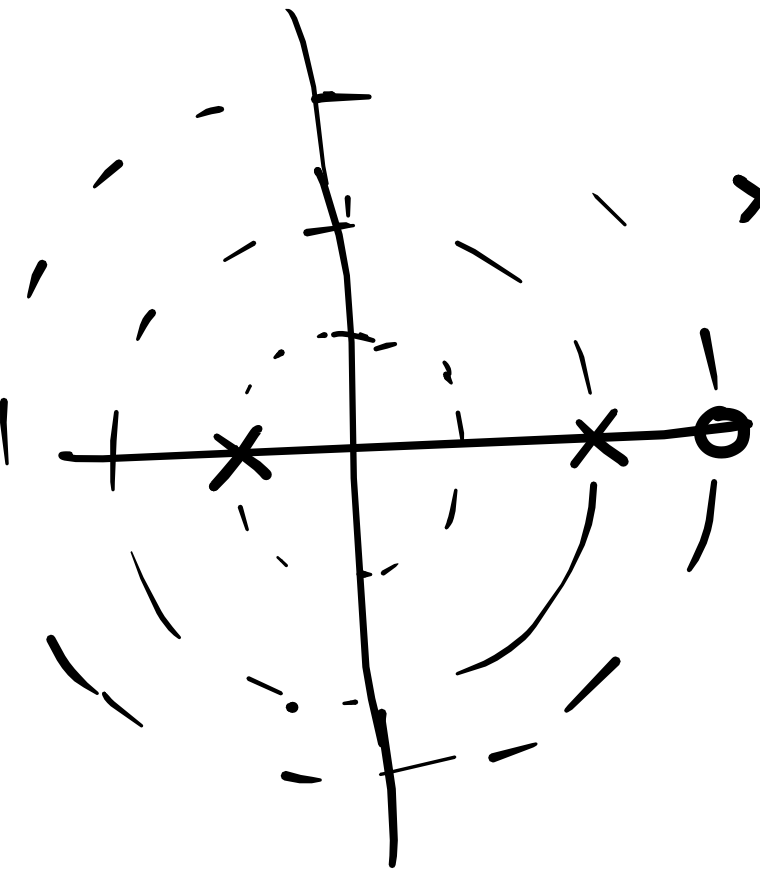
In discrete, frequencies are unique only on 2π interval.



For $X(z)$, which values of z make $X(z)$ finite-valued $\Rightarrow$ Region of convergence ROC

$$X(z) = \text{Rational fn. in } z = \frac{N(z)}{D(z)}$$

Let $$X(z) = \frac{(z-3)}{(z+1)(z-2)}$$



$X(z)$ has a zero at $z=3$

has poles at $z=-1$ & $z=2$

No. of poles = no. of zeros.

a zero is sitting at $z=\infty$.

Ex

$$x[n] = a^n u[n]$$

$$X(z) = \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1-az^{-1}} \quad \text{if } |az^{-1}| < 1$$

$$\text{if } |a| < |z| \quad \text{or} \quad |z| > |a|$$

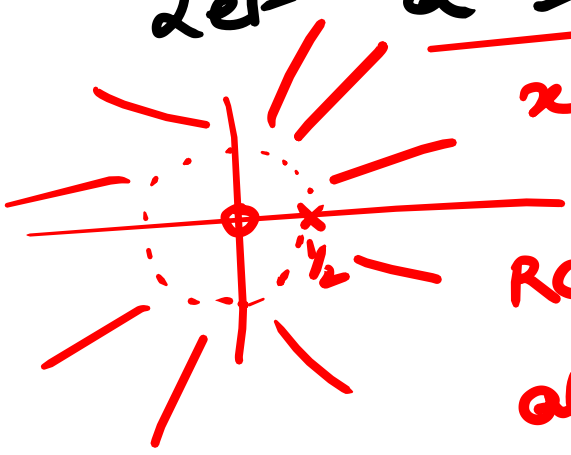
ROC

$z = re^{j\omega}$

$= |z| e^{j\omega}$

Let $a = 0.5$

$x[n] = \left(\frac{1}{2}\right)^n u[n] = \left(\frac{z}{z-a}\right)$



at Unit circle $z = e^{j\omega}$, & this is included in ROC $\Rightarrow$ $x[n]$ is F.T transformable.

Let $a = 2$

$|z| > 2$ $\Rightarrow$ does not include circle, F.T. does not exist. ✗

$x[n] = 2^n u[n]$

$\Downarrow$

Not absolutely summable..

Ex $x[n] = -a^n u[-n-1]$

$$X(z) = -\sum_{n=-\infty}^{-1} a^n z^{-n} = -\sum_{n=1}^{\infty} (a^{-1}z)^n - 1 + 1$$

$$= -\sum_{n=0}^{\infty} (a^{-1}z)^n + 1 = 1 - \frac{1}{1-a^{-1}z}$$

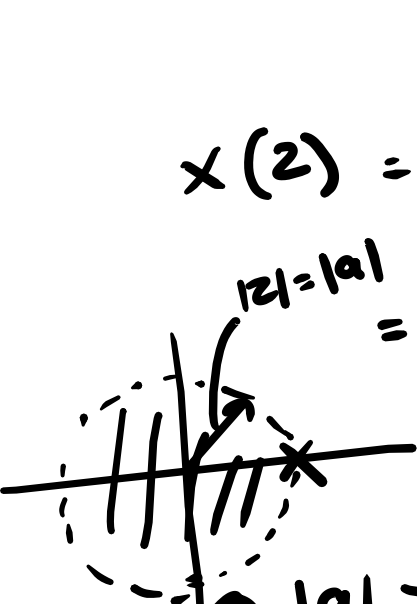
where $|a^{-1}z| < 1$

$|z| < |a|$

$$X(z) = 1 - \frac{a}{a-z}$$

$$= \frac{a-z-a}{a-z} = \frac{-z}{a-z} \qquad |z|=|a|$$

$$= \frac{z}{z-a} \qquad |z| < |a|$$



① $|a| > 1$ — F.T exists

② $|a| = 1$ — F.T does not exists

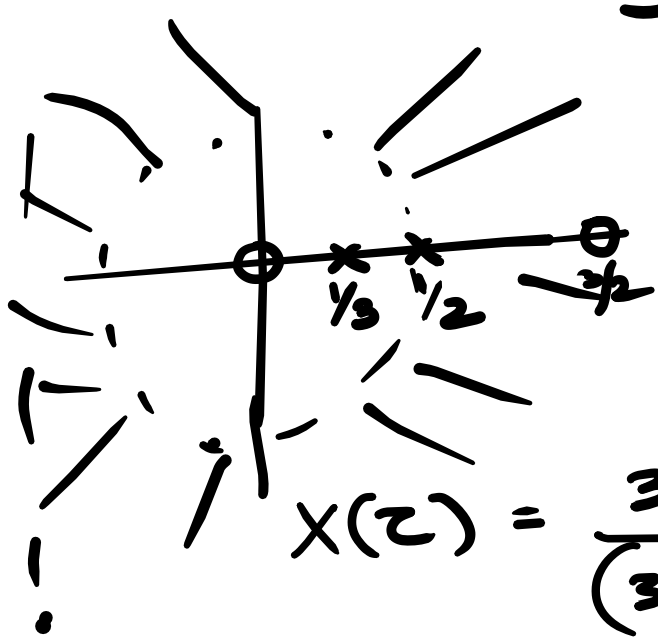
③ $|a| < 1$ — F.T does not exist

Ex $x[n] = 7\left(\frac{1}{3}\right)^n u[n] - 6\left(\frac{1}{2}\right)^n u[n]$

$$X(z) = \frac{7}{1-\frac{1}{3}z^{-1}} - \frac{6}{1-\frac{1}{2}z^{-1}}$$

$\Downarrow$ $\Downarrow$

ROC $= |z| > \frac{1}{3}$ $\quad |z| > \frac{1}{2}$



ROC is outside the outermost pole

$$X(z) = \frac{32(2z-3)}{(3z-1)(2z-1)}$$

$$= \frac{z(z-3/2)}{(z-1/3)(z-1/2)}$$

Ex $x[n] = \left[\left(\frac{1}{3}\right)^n \sin\left(\frac{\pi}{4}n\right)\right] u[n]$

$$X(z) = \frac{1}{j}\sum_{n=0}^{\infty}\left(\frac{e^{j\frac{\pi}{4}n} - e^{-j\frac{\pi}{4}n}}{2}\right) z^{-n}$$

$$= \frac{1}{18j}\left\{\sum_{n=0}^{\infty}\left(e^{j\frac{\pi}{4}}z^{-1}\right)^n - \sum_{n=0}^{\infty}\left(e^{-j\frac{\pi}{4}}z^{-1}\right)^n\right\}$$

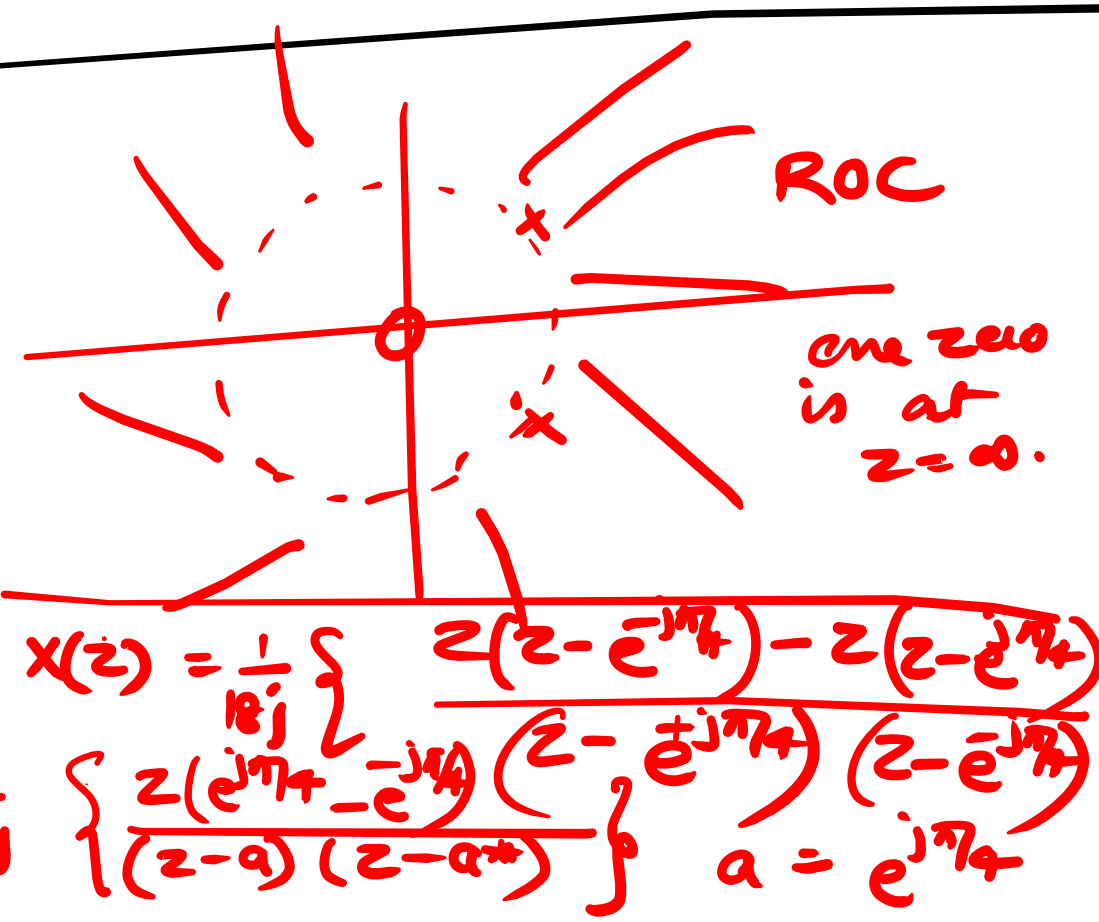
$$= \frac{1}{18j}\left\{\frac{1}{1-e^{j\frac{\pi}{4}}z^{-1}} - \frac{1}{1-e^{-j\frac{\pi}{4}}z^{-1}}\right\}$$

$\Downarrow$ $|z| > 1$

$\Downarrow$ $|z| > 1$

$\Downarrow$ Simplify

$$x[n] = \left(\left(\frac{1}{3}\right)^n \sin\frac{\pi}{4}n\right) u[n]$$

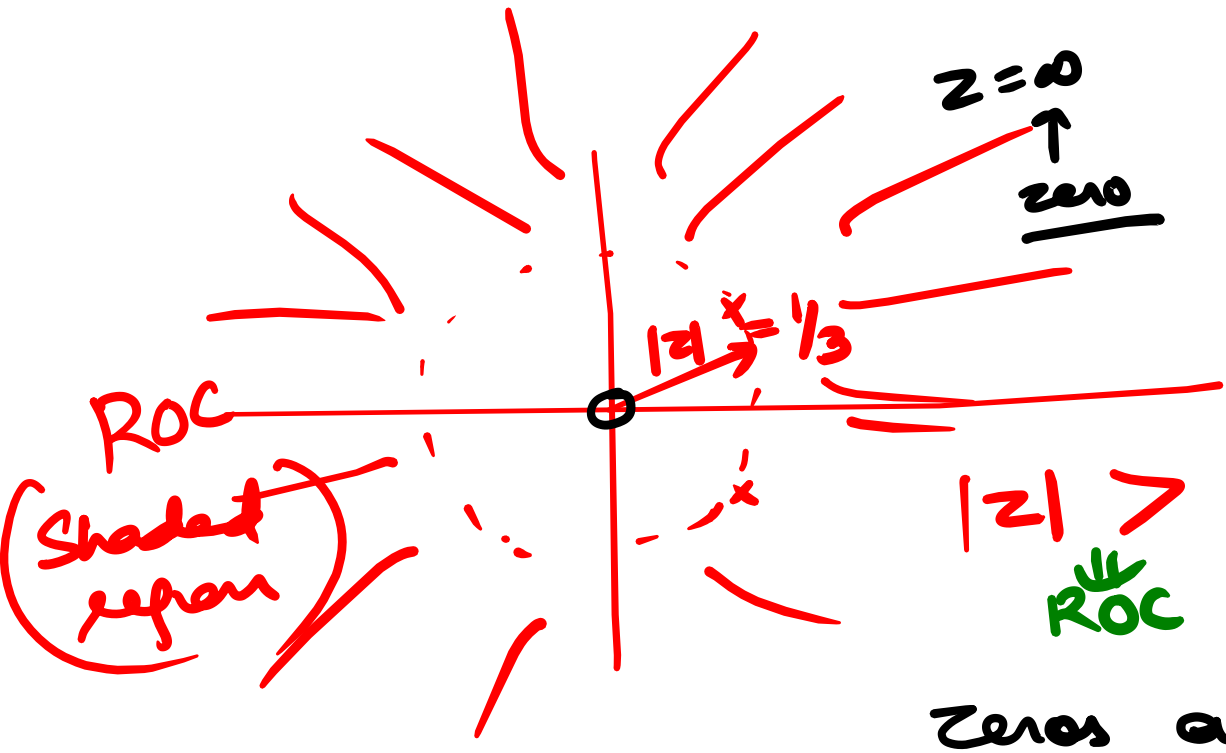


$$X(z) = \frac{1}{18j}\left\{\frac{z(z-e^{-j\frac{\pi}{4}}) - z(z-e^{j\frac{\pi}{4}})}{(z-e^{j\frac{\pi}{4}})(z-e^{-j\frac{\pi}{4}})}\right\}$$

$$= \frac{1}{18j}\left\{\frac{z(e^{j\frac{\pi}{4}} - e^{-j\frac{\pi}{4}})}{(z-a)(z-a^*)}\right\} \quad a = e^{j\frac{\pi}{4}}$$

Ex $x[n] = \left(\frac{1}{3}\right)^n \sin\left(\frac{\pi}{4}n\right) u[n]$

$$= \frac{1}{2j}\sum_{n=0}^{\infty}\left(\frac{1}{3}e^{j\pi/4}z^{-1}\right)^n - \frac{1}{2j}\sum_{n=0}^{\infty}\left(\frac{1}{3}e^{-j\pi/4}z^{-1}\right)^n$$



poles are at
$z = a$ & $z = a^*$
where $a = \frac{1}{3}e^{j\pi/4}$

$x[n]$ is Fourier transformable
$\therefore$ ROC includes the unit circle

Zeros are at $z=0$ & $z=\infty$

Say, it is an LTI system → IIR system

Ex $h[n] = a^n u[n]$ $a > 0$

$H(z) = \frac{1}{1-az^{-1}}$ $|z| > |a|$

Case-1: $a \geq 1$

Case-2: $0 < a < 1$

Case-1 : ROC does not include unit circle
⇒ The system is not stable.

Case-2 $|z| > |a|$ ⇓ ROC includes unit circle

⇒ F.T. exists

⇒ $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$ Sequence is absolutely summable

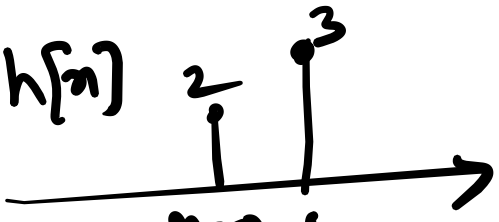
⇓

⇒ requirement for BIBO stability ⇒ Hence, The system is stable.

LTI system

Ex $h[n] = 2\delta[n] + 3\delta[n-1]$

Find the z-transform
Plot poles & zeros
Plot ROC
Comment on stability & causality



FIR system

An FIR system is always stable ∵ it is non-zero only over the finite duration.

$x[n]\delta[n-n_0] = x[n_0]\delta[n-n_0]$

Ex $x[n] = b^{|n|}$

Case-1 : $0 < b < 1$

Case-2 : $b \geq 1$

Find the z-transform.
Plot ROC, poles & zeros.

$$H(z) = \sum_{n=-\infty}^{\infty} h[n] z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} (2\delta[n] + 3\delta[n-1]) z^{-n}$$

$$= 2\delta[n] z^{0} + 3\delta[n-1] z^{-1}$$

$$= 2 + 3z^{-1}$$

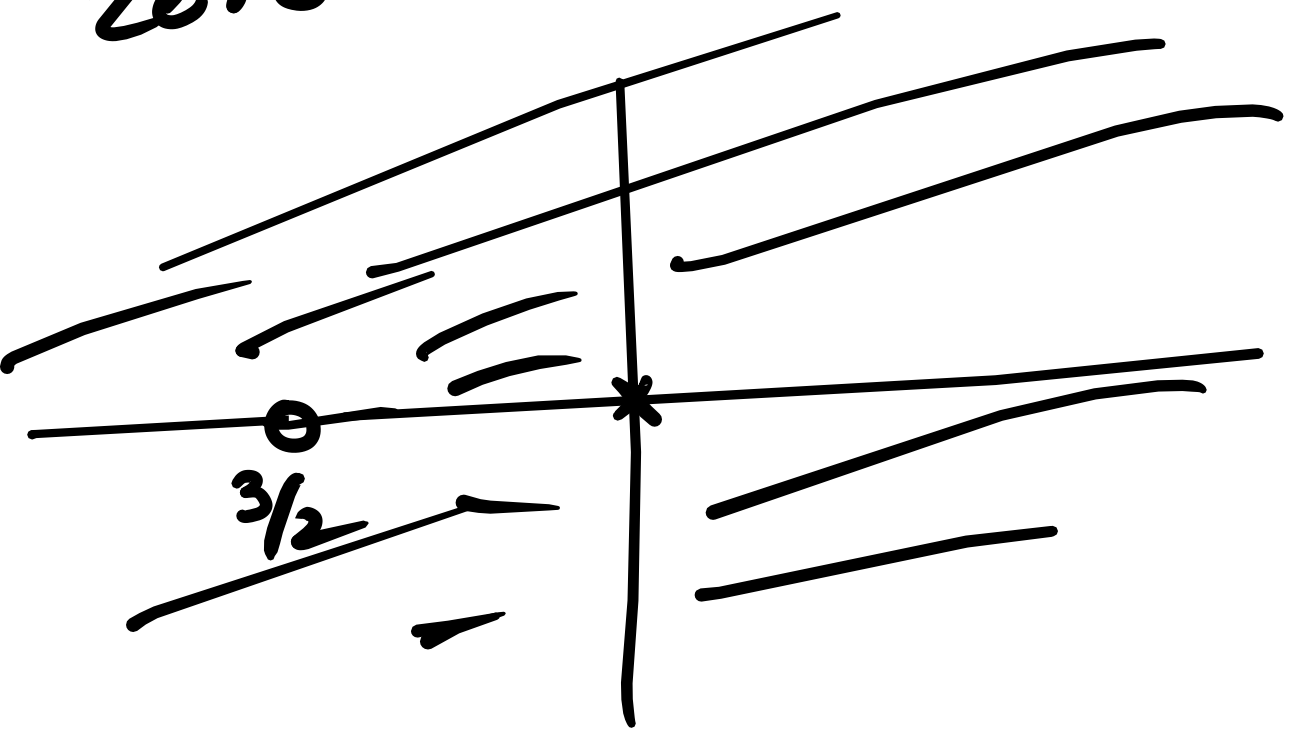
$x[n]\delta[n] = x[0]\delta[n]$

$$H(z) = 2 + 3z^{-1}$$

$$= \frac{2z + 3}{z}$$

pole at $z = 0$

zero at $z = -3/2$



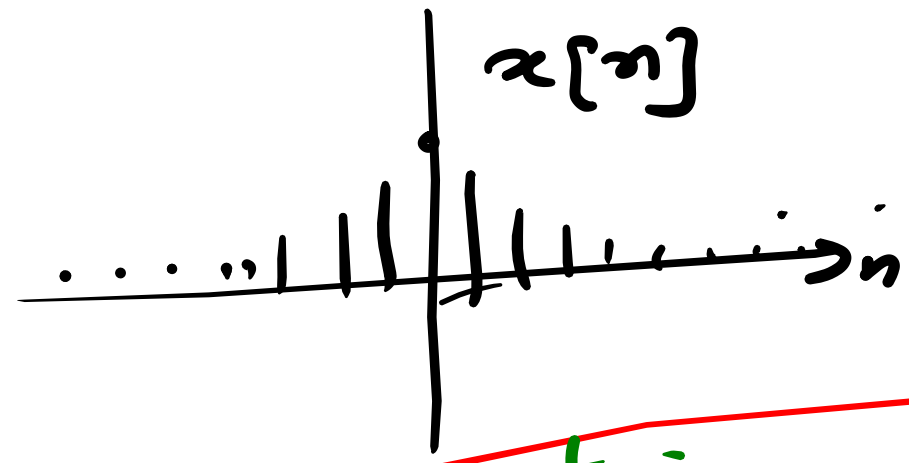
ROC = Entire z-plane excluding $z = 0$

The system is causal & stable.

ROC excludes $z = 0$

Ex $x[n] = b^{|n|}$

Case 1 : $0 < b < 1$



$$x[n] = \begin{cases} b^n & n \geq 0 \\ b^{-n} & n < 0 \end{cases}$$

$$X(z) = \sum_{n=-\infty}^{-1} b^{-n} z^{-n} + \sum_{n=0}^{\infty} b^n z^{-n}$$

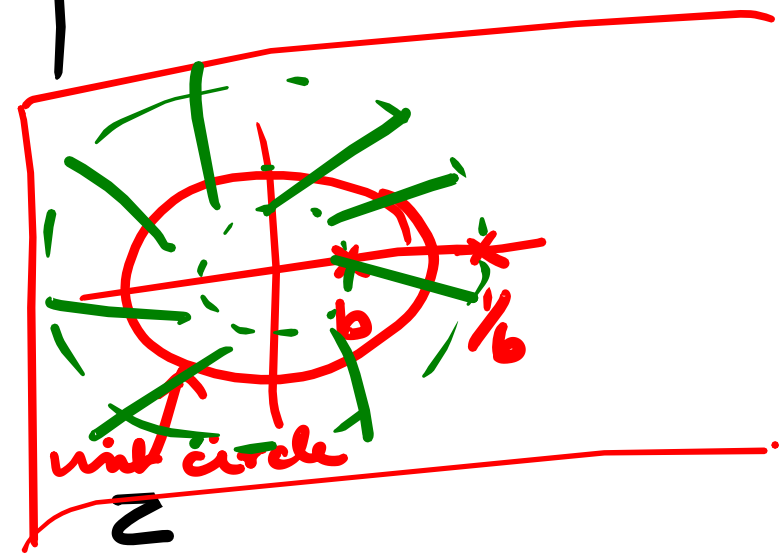
ROC: $|z| < b$

$$= \frac{bz}{1-bz} + \frac{z}{z-b} = \frac{z}{(z-b)} - \frac{z}{z-1/b}$$

$|z| < \frac{1}{b}$

$$= \frac{z(b-1/b)}{(z-b)(z-1/b)}$$

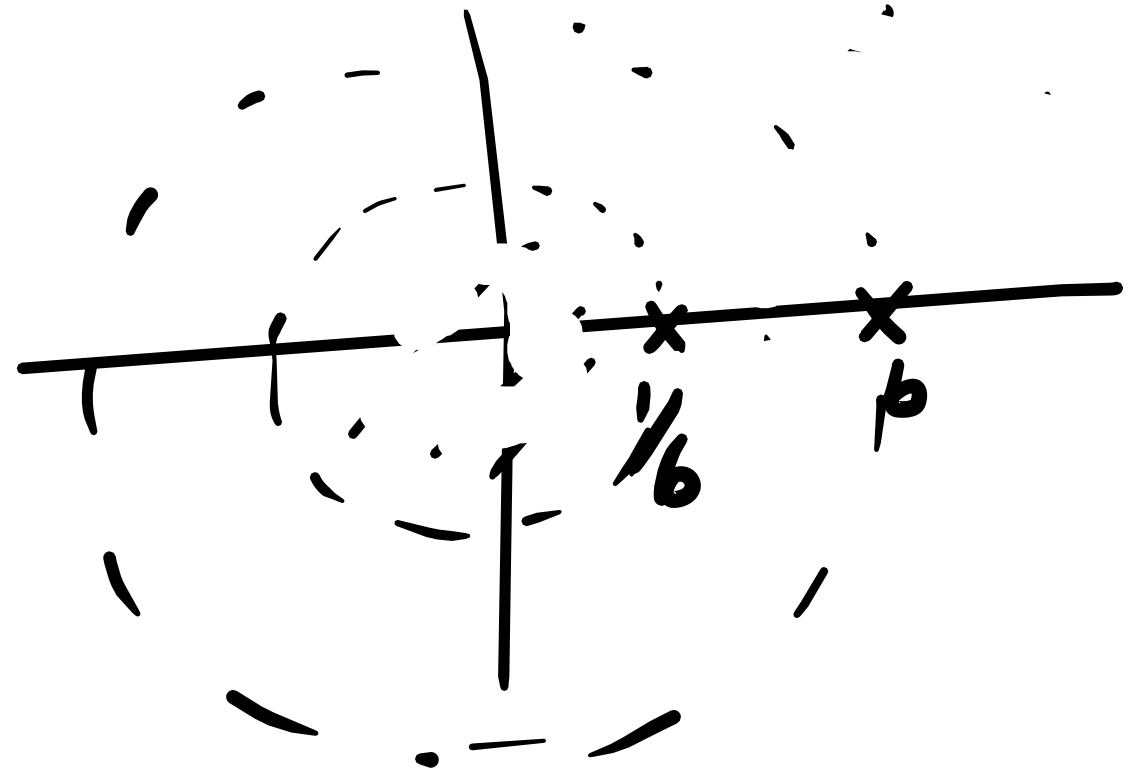
ROC: $b < |z| < \frac{1}{b}$



Case-2 $b > 1$

$$X(z) = \frac{bz}{1-bz} + \frac{z}{z-b}$$

ROC: $|z| < \frac{1}{b}$ and $|z| > b$



there is no common region of convergence

Hence, Z-transform does not exist.

Ex Find the signal $x[n]$, given that

$$X(z) = \frac{3 - \frac{9}{6}z^{-1}}{\left(1 - \frac{1}{4}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)} \quad ; \text{ ROC : } |z| > \frac{1}{3}$$

$$= \frac{A}{1 - \frac{1}{4}z^{-1}} + \frac{B}{1 - \frac{1}{3}z^{-1}}$$

$$= \frac{1}{1 - \frac{1}{4}z^{-1}} + \frac{2}{1 - \frac{1}{3}z^{-1}}$$

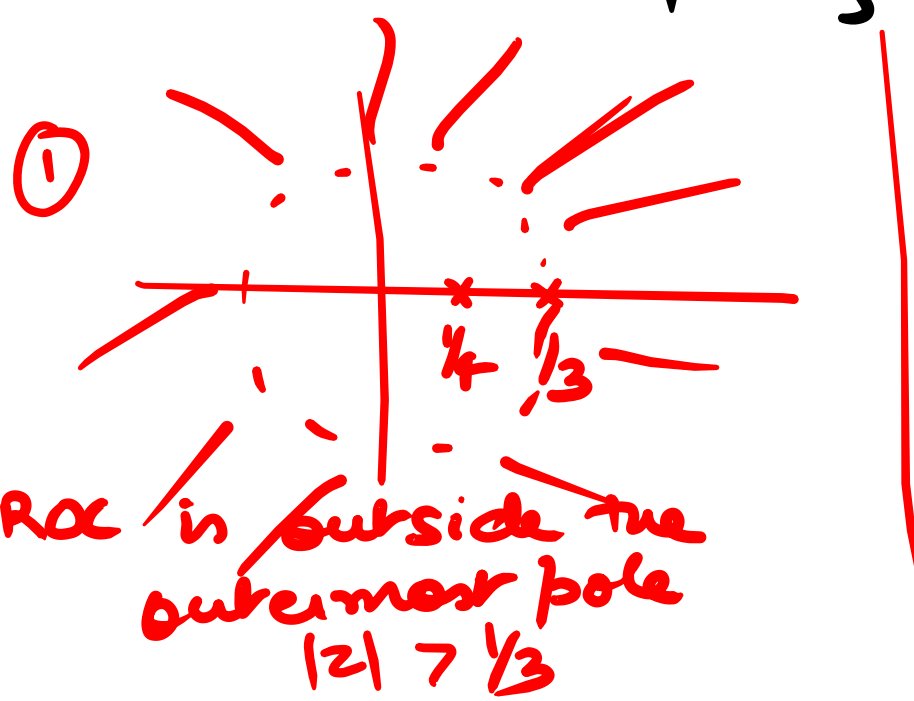
$|z| > \frac{1}{4}$ ⇓ ∩ ⇓ $|z| > \frac{1}{3}$

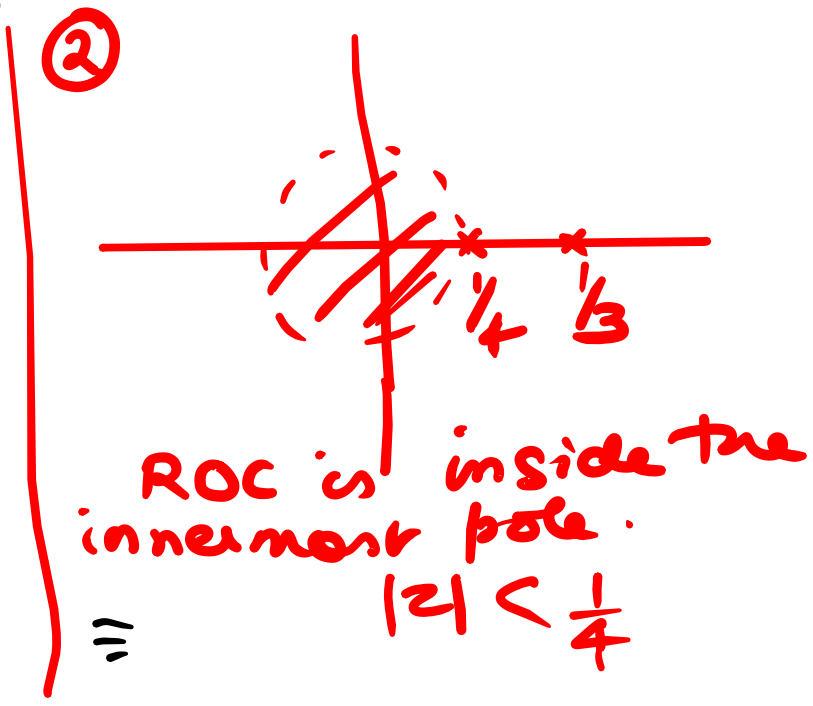
intersection $|z| > 1/3$

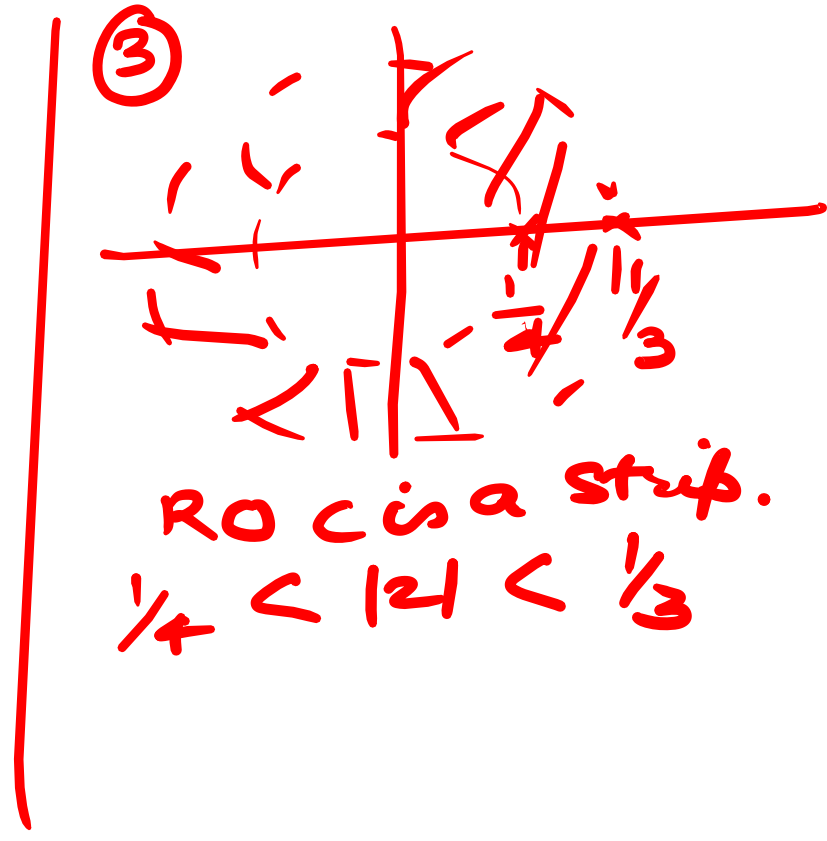
$$\therefore x[n] = \left(\frac{1}{4}\right)^n u[n] + 2\left(\frac{1}{3}\right)^n u[n]$$

$$X(z) = \frac{3 - \frac{5}{6}z^{-1}}{\left(1 - \frac{1}{4}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)}$$

Draw all ponible ROC & determine $x[n]$ corresponding to each.







$$X(z) = \frac{1}{1-\frac{1}{4}z^{-1}} + \frac{2}{1-\frac{1}{3}z^{-1}}$$

Case-1

ROC : $|z| > \frac{1}{3}$

$|z| > \frac{1}{4}$ $\quad$ $|z| > \frac{1}{3}$

$$\Rightarrow x[n] = \left(\frac{1}{4}\right)^n u[n] + 2\left(\frac{1}{3}\right)^n u[n]$$

Case-2 ROC : $|z| < \frac{1}{4}$

$\frac{1}{1-\frac{1}{4}z^{-1}}$; $|z| < \frac{1}{4}$ | $\frac{2}{1-\frac{1}{3}z^{-1}}$; $|z| < \frac{1}{3}$

$x[n] = -\left(\frac{1}{4}\right)^n u[-n-1] - 2\left(\frac{1}{3}\right)^n u[-n-1]$

$\Rightarrow$ $\Lleftarrow$

$x_1[n] = a^n u[n]$

$x_2[n] = -a^n u[-n-1]$

Case-3

$\frac{1}{4} < |z| < \frac{1}{3}$

$x[n] = \left(\frac{1}{4}\right)^n u[n]$

$- 2\left(\frac{1}{3}\right)^n u[-n-1]$

Ex

$$x[n] = \begin{cases} a^n & ; \quad 0 \le n \le N-1, \; a > 0 \\ 0 & ; \quad \text{otherwise} \end{cases}$$

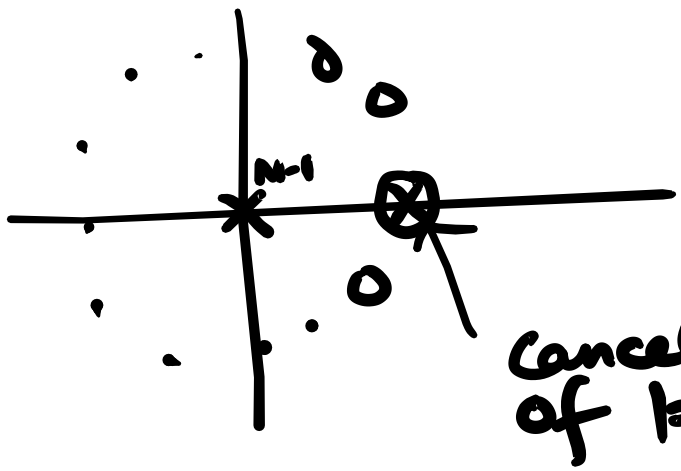
$$X(z) = \sum_{n=0}^{N-1} (az^{-1})^n = \frac{1-(az^{-1})^N}{1-(az^{-1})}$$

ROC is entire z-plane excluding $z = 0$

$$= \frac{1}{z^{N-1}} \left(\frac{z^N - a^N}{z - a} \right)$$

zeros are at $z_k = ae^{j\frac{2\pi}{N}k}$

$k = 0, \ldots N-1$



cancellation of pole & zero at $z = a$

$x[n]$ is real,

$X(z)$ has poles & zeros in conjugate pairs if they are complex.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$X^*(z) = \sum_{n=-\infty}^{\infty} x^*[n] (z^*)^{-n}$$

$$X^*(z^*) = \sum_{n=-\infty}^{\infty} x^*[n] z^{-n}$$

If $x[n]$ is real, $x[n] = x^*[n]$

$$\therefore \quad X^*(z^*) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = X(z)$$

if signal is real,

$X(z) = X^*(z^*)$

↓ ↓

if there is a root of $N(z)$ or $D(z)$ at $z=a$, there is also a root at $z=a^*$ or $z^*=a$

10.17

S: LTI system

(1) $h[n]$ is real

(2) $h[n]$ is right-sided

(3) $\lim_{z \to \infty} H(z) = 1$

(4) $H(z)$ has two zeros

(5) $H(z)$ has one of the pole at nonreal location on the circle defined by $|z| = 3/4$

Answer

(a) is S causal?

(b) Is S stable?

System is causal & Stable

$$H(z) = \frac{A(z - z_0)(z - z_0^*)}{\left(z - \frac{3}{4}e^{j\phi}\right)\left(z - \frac{3}{4}e^{-j\phi}\right)}$$

↑ ① , ④ , ⑤

$$H(z) = \sum_{n=-\infty}^{\infty} h[n] z^{-n}$$

$\lim_{z \to \infty} H(z) = 1$

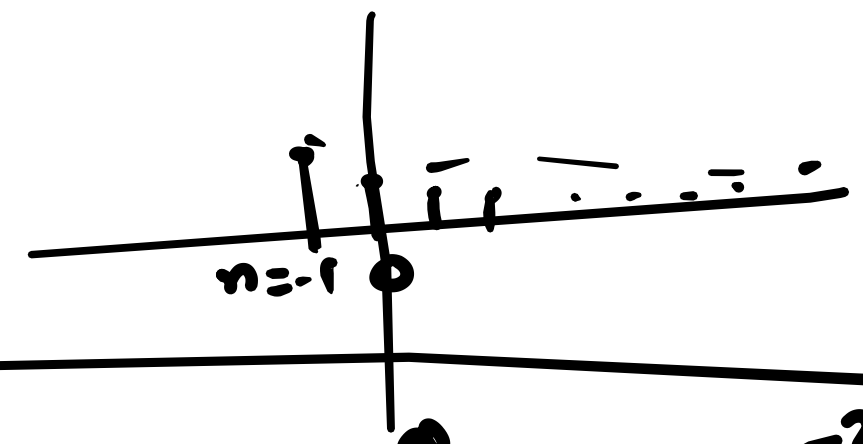
⇒ Divide $N(z)$ & $D(z)$ by z^2 & put $\lim z \to \infty$

$A = 1$

$$h[n] = a^n u[n+1] \quad ; 0<a<1$$

$h[n] \neq 0 \qquad n<0$

$$= \frac{1}{a}\left(a^{n+1}u[n+1]\right)$$



$$x[n] \rightleftharpoons X(z)$$

$$x[n+1] \rightleftharpoons zX(z)$$

$$X(z) = \sum_{n=-0}^{\infty} x[n]z^{-n}$$

$$Y(z) = \sum_{n=-0}^{\infty} x[n+1]z^{-n}$$

$$= \sum_{n=-0}^{\infty} x[n]z^{-(n-1)}$$

$m = n+1$